

Texas College of Management & IT

Bachelor of Computer Science (Hons) (Network Technology & Cybersecurity)

Mini Project

Title: Practical Implementation of a Database-Driven Student Information Management System

Academic Year: 2026 (4th Semester)

Course: Database Management Systems

Subject Code: (DMS - 236)

Purpose

To evaluate students' practical skills in designing, implementing, and managing relational databases using SQL and DBMS concepts.

Learning Outcomes (LOs)

Students will be able to:

- Design conceptual and logical database models (ER diagrams, normalization).
- Implement relational schema with constraints and keys.
- Perform CRUD operations using SQL.
- Apply relational algebra queries.
- Manage transactions.

Objectives

- Assess hands-on database design and implementation skills.
- Evaluate practical application of DBMS concepts in real-world scenarios.
- Strengthen query writing, debugging, and testing abilities.

Project Scenario

Develop a **Student Information Management System** that includes:

- Student registration and profile management.
- Course enrollment and grade storage.
- Faculty feedback and evaluation records.
- Admin monitoring of student and faculty data.

Task: Practical Components (100 Marks)

Perform ALL tasks and include screenshots, source code, and explanations.

Practical 1: Conceptual & Logical Design (20 Marks)

- Draw ER Diagram for the system.
- Normalize tables up to 3NF.
- Include:
 - ER diagram.
 - Normalization (1,2,3NF).

Practical 2: Database Creation (20 Marks)

- Create tables in MsSQL:
 - Students
 - Courses
 - Enrollment
 - Faculty
 - Feedback
- Define primary keys, foreign keys, and constraints.
- Include:
 - SQL scripts.
 - Database screenshots.

Practical 3: CRUD Operations & Queries (20 Marks)

- Implement SQL queries for:
 - Insert, Update, Delete student/course records.
 - Retrieve student details with joins.
 - Search using conditions (LIKE, BETWEEN, IN).
- Include:
 - SQL queries.
 - Output screenshots.

Practical 4: Relational Algebra & Transactions (20 Marks)

- Demonstrate relational algebra queries (Select, Project, Join, Union, Intersection).
- Implement transaction management with COMMIT, ROLLBACK.
- Include:
 - Relational algebra examples.
 - Transaction test cases.

Practical 5: Database Operations (20 Marks)

- SQL Joins
 - Inner Join: Retrieve student details with their enrolled courses.
 - Left/Right Join: Show students who have not enrolled in any course.
 - Self Join: Compare students' grades within the same course.
- Aggregate Functions
 - COUNT: Total number of students enrolled in each course.
 - SUM: Calculate total credits earned by a student.
 - AVG: Find average grade per course.
 - MAX/MIN: Identify highest and lowest grades in a subject.
- Grouping & Filtering
 - GROUP BY: Generate reports of students grouped by course.
 - HAVING: Filter courses with average grade below a threshold.

Evaluation and Marking Scheme

Criteria	Description	Weight
Functionality	Working features	30%
Database Design	ER diagram & normalization	20%
Query	SQL & relational algebra	20%
Documentation	Clarity & completeness	15%
Output	Screenshots	15%

Submission Requirements

Students must submit:

- Complete SQL scripts
- Screenshots of queries and outputs
- ER diagram and normalization steps
- Short Report
- Presentation slides

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**Project Title: *Practical Implementation of a Database-Driven Student
Information Management System***

Submitted By:

Name: _____

LCID: _____

Year/Semester: BCS 2nd Year / 4th Semester

Submitted To:

Lecturer Name: Dr.Sushil Bhattarai

Course: Database Management Systems((DMS - 236)

Date of Submission: _____

Texas College of Management & IT, Kathmandu

Assignment Structure

1. Title Page
2. Table of Contents
3. Introduction
4. Task Implementation
5. Conclusion
6. References (APA 7th edition)
7. Appendix (Screenshots, Scripts, Logs)

Formatting Guidelines

- **Page Size:** A4
- **Margins:** 1 inch (left 1.5 inch)
- **Font:** Times New Roman
- **Font Size:**
 - Body: 12 pt
 - Headings: 14–16 pt
- **Line Spacing:** 1.5 or double
- **Page Numbering:**
 - Roman (i, ii) for preliminary pages
 - Arabic (1, 2) for main body
- **Referencing Style:** APA (Latest Edition)